



Pralay Kavach

a shield against the deluge

AI cyclone early warning and anticipatory action for India's east coast: from the first forecast to families in shelters.

Build with AI: Code for Communities

Resilience track · Gemini · Earth Engine · Cloud Run



THE PROBLEM

Forecasts got better. *The last mile didn't.*

Cyclone Fani, Odisha, 3 May 2019

185 km/h

at landfall near Puri: an extremely severe cyclonic storm

~ 12 lakh

people evacuated by the Odisha government, a massive, largely manual effort

1

Who must move?

Surge, wind and population are separate maps; officers merge them by hand under time pressure.

2

Warn everyone in their language

Advisories in Odia, Telugu, Bengali and English, written, checked and approved at 2 a.m.

3

Reach people, not just phones

Residents need their own shelter, a safe route and someone to ask, in their language.

4

Where did the water actually go?

Relief and insurance need the real flood footprint, days before surveys finish.

THE SOLUTION

One flow from forecast to *shelter*.



01

Watch

Live wind, satellites and a forming-storm detector over the Bay, every day.



02

Assess

Storm surge, wind and waterlogging on real terrain: who and what is exposed.



03

Plan

A Gemini agent reads the bulletin and satellite image, ranks villages and explains why.



04

Warn

Advisories in four languages, two-person approval, CAP alerts, a voice helper on every phone.



05

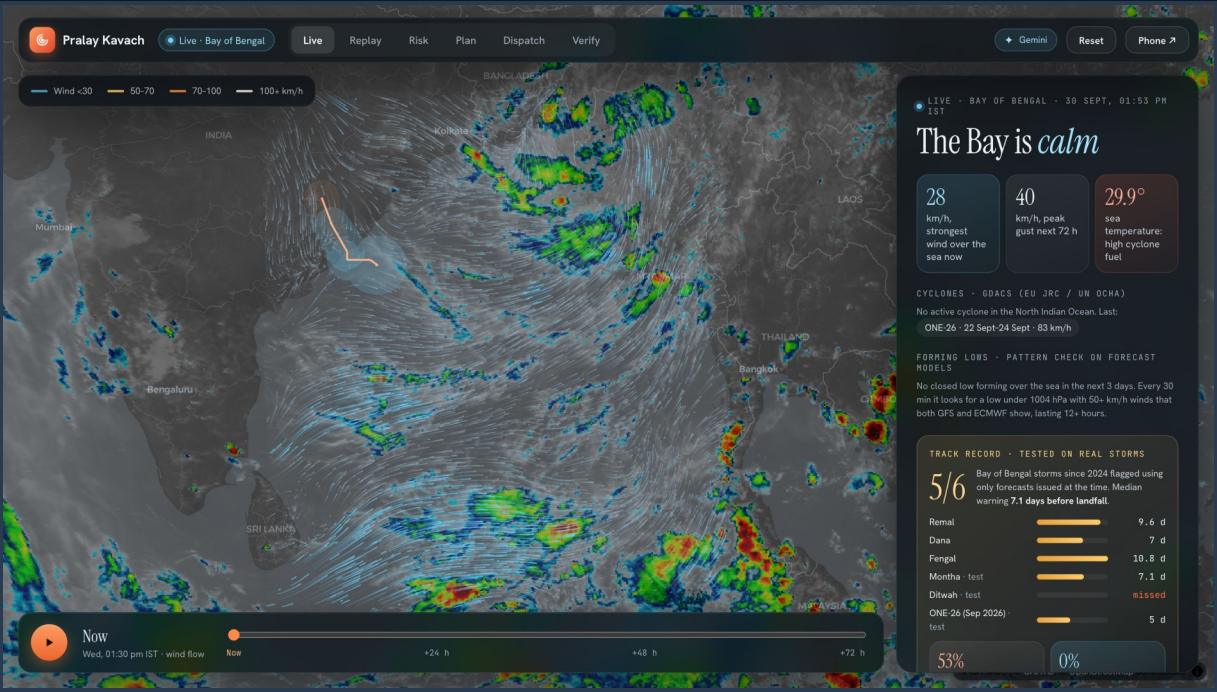
Verify

Sentinel-1 radar shows where water stood; relief goes there first; parametric payouts trigger.

Gemini powers Plan, Warn (advisories, voice helper) and photo checks. Earth Engine powers Assess and Verify.

WATCH: EVERY DAY, NOT JUST IN A DISASTER

Spots forming storms a week out, *without crying wolf.*



Live: GFS wind flow, Himawari clouds, GPM rain, GDACS storms and a plain-language outlook for any place.

0 of 7

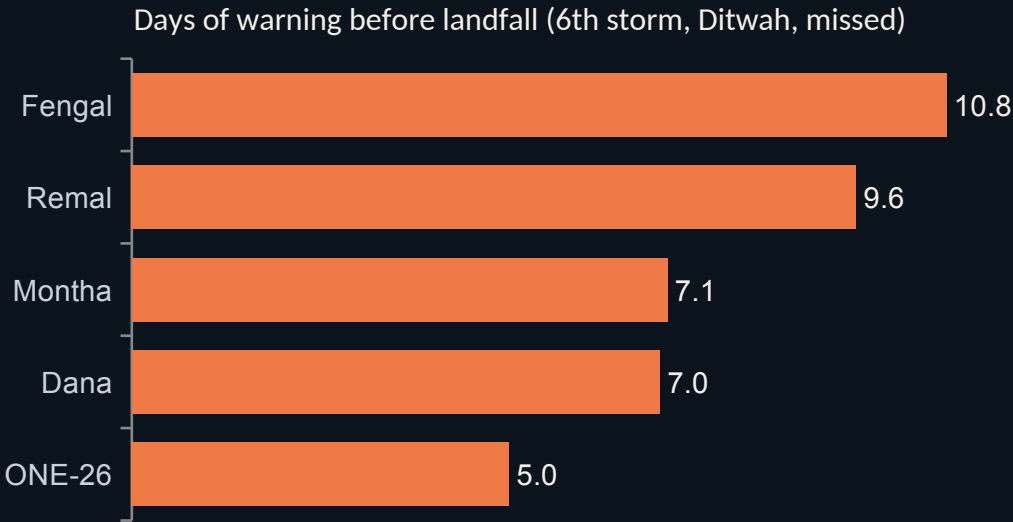
5 of 6

7.1 d

false alarms on held-out 2025–26 (first version: 18 of 34)

storms spotted before landfall

median warning lead



Who moves first, *and why*.

Reads the cyclone bulletin and the satellite image together (Gemini multimodal).

Runs surge on three track scenarios, wind exposure, shelters and road cut-off timing.

Ranks villages by people at risk and leave-by time, with a reason for each.

Logs all six steps on screen; it may only use numbers the models produced.

Never breaks if the model is slow or down, rules take over and the screen says so.

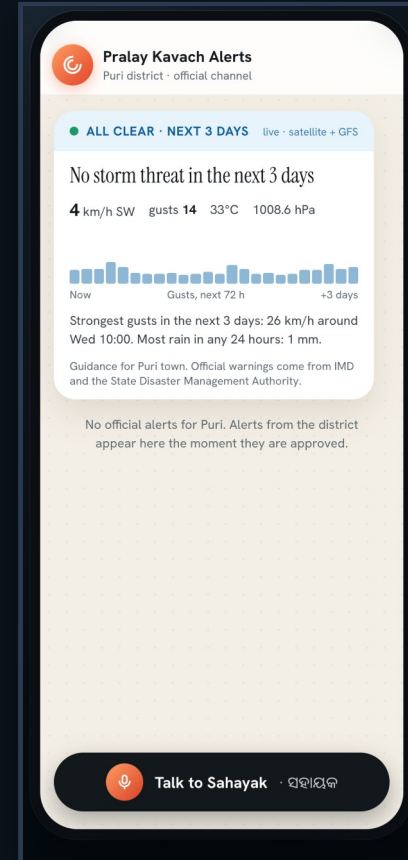


WARN: IN EVERY LANGUAGE, TO EVERY PHONE

From approval to a ringing phone in *seconds*.

In the control room

- 1 Draft** Gemini writes the advisory in Odia, Telugu, Bengali and English, action first, under 60 words, only from the model facts.
- 2 Check** Each language comes with an English back-translation so the officer can verify it.
- 3 Approve** Duty officer, then Relief Commissioner: two-person sign-off.
- 4 Send** A standard CAP 1.2 alert; the phone rings within seconds.



Sahayak

voice helper on the resident's phone

6 languages, spoken aloud

Grounded on the person's own alert, shelter and route

Acts: files SOS and water reports that reroute others

Safety in code: "Call 112" first on danger words

VERIFY: CHECKED AGAINST CYCLONE FANI

Where water *actually* stood.

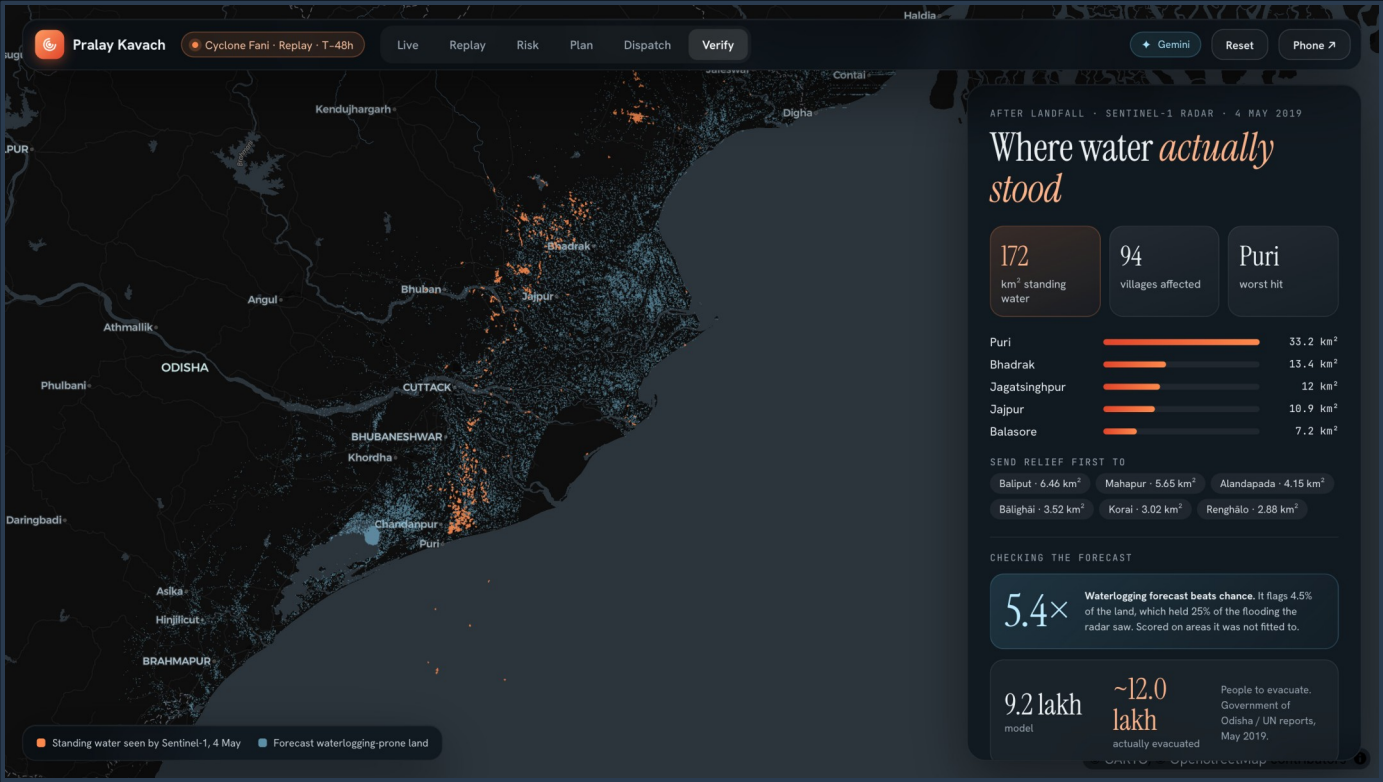
5.4×

better than chance: waterlogging forecast vs Sentinel-1 radar, on areas it was not fitted to

~9 vs ~12 lakh

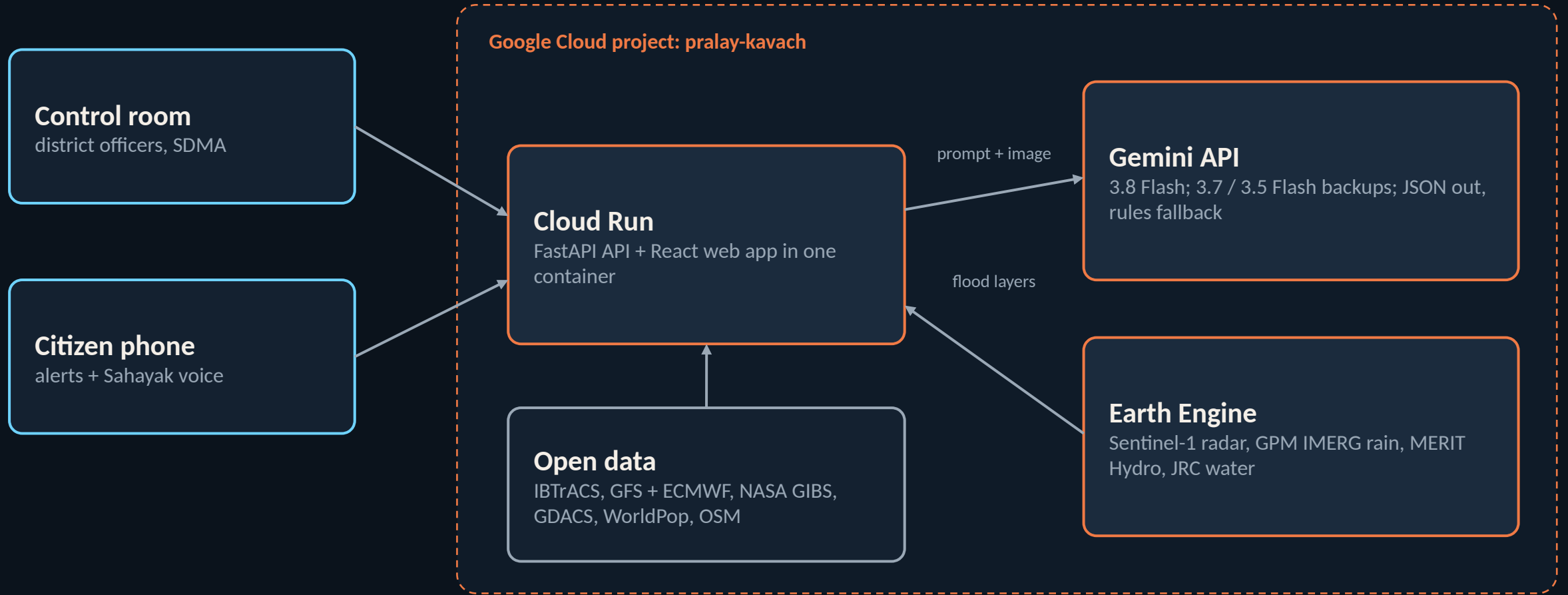
people to evacuate: our model vs what Odisha actually moved

Honest about limits. Surge drained before the radar pass, so it is not scored.



ARCHITECTURE

Built on *Google Cloud and Gemini.*



Gemini: agent planning (multimodal) · multilingual advisories · Sahayak voice helper · flood photo checks

Built for one coast, *ready for others.*



Residents

A call in their language, their own shelter and route, and someone to ask for help.



District control rooms

Hours saved on who-moves-first, advisories and sign-off, with every step logged.



Relief and insurers

Radar-based relief lists and parametric payouts days before ground surveys.

Same pipeline, other coasts

India: Andhra Pradesh, West Bengal Sundarbans, Gujarat

Mozambique: Idai, Freddy (Portuguese)

South Africa: KwaZulu-Natal coast (Zulu, English)

China: Guangdong typhoon coast (Mandarin)

Brazil: southern coast, Catarina (Portuguese)

Inputs are global: IBTrACS, GFS/ECMWF, Copernicus, NASA, WorldPop, OSM.
Change the map extent and language list.

WHAT'S NEXT

Pilot before the next *cyclone season*.



Now

Working prototype: live Bay watch, Fani replay, Gemini agent, voice helper, radar checks.

Next 3 months

Official shelter registry (OSDMA), Google Maps Routes for real road routing, IVR and WhatsApp on the same Sahayak engine.

Pilot

One Odisha coastal district, run alongside IMD and SDMA warnings during the season.

Code github.com/nsquaredzz/Hackathons (folder 2026-09-pralay-kavach-cyclone-early-warning) *Guidance only: official warnings come from IMD and the SDMA.*